Banking Loan Risk Analysis Application

Dataset Details, Schema Dictionary & Preprocessing Guide

1. Dataset Overview

- **Dataset Name:** Default of Credit Card Clients Dataset
- **Source:** UCI Machine Learning Repository / Chung Hua University & Tamkang University, Taiwan
- **Format:** Microsoft Excel (.xls) / CSV (.csv)
- **Record Count:** 30,000 customers
- **Target Variable:** default payment next month (Binary: 1 = default, 0 = non-default)

2. Attribute Definitions

- **ID:** Customer identifier (1 to 30,000).
- **LIMIT_BAL:** Amount of given credit line in NT Dollars (NT0,000 to NT,000,000; mean NT67,484).
- **SEX:** Gender (1 = Male [11,888], 2 = Female [18,112]).
- **EDUCATION:** Education level (1 = Graduate School [10,585], 2 = University [14,030], 3 = High School [4,917], 4 = Others [123], 0/5/6 = Unknown [345]).
- **MARRIAGE:** Marital status (1 = Married [13,659], 2 = Single [15,964], 3 = Others [323], 0 = Unknown [54]).
- **AGE:** Age in years (Range: 21 to 79 years; mean ~35.5 years).
- **PAY_0 to PAY_6:** Historical repayment statuses for 6 consecutive months (September 2005 to April 2005). Scale: -2 = no consumption, -1 = paid duly, 0 = revolving credit, 1 to 8 = months of delay.
- **BILL_AMT1 to BILL_AMT6:** Bill statement balances across the 6 billing cycles.
- **PAY_AMT1 to PAY_AMT6:** Actual cash payments made across the 6 billing cycles.

3. Preprocessing & Feature Engineering

- Cleaned header rows and standardized column identifiers.
- Decoded numeric codes to human-readable labels (e.g. Male, Female, Graduate School, Married).
- Derived **age_group** brackets: 20-29, 30-39, 40-49, 50-59, 60+.
- Engineered **utilization_ratio** (average bill / credit limit).
- Engineered **repayment_ratio** (total payments / total bills).
- Engineered **max_delay_months** and delinquent cycles count.